Question 3

(10 marks)

A beam-splitter is an optical device with two input modes and two output modes. Consider a 50:50 beam splitter with input modes 1 and 2 and output modes 3 and 4. This beam-splitter combines the modes as $\hat{a}_1 = (\hat{a}_3 + \hat{a}_4)/\sqrt{2}$ and $\hat{a}_2 = (\hat{a}_3 - \hat{a}_4)/\sqrt{2}$.

(a) (3 marks) What output state is expected when the two inputs each have a single photon in them $|1\rangle_1|1\rangle_2$?

(b) (2 marks) Show that for any input this beam-splitter conserves energy.

Consider the normalised state $|\psi\rangle = \alpha|n\rangle + \beta|n+1\rangle$ where $|n\rangle$ is the n-photon state and $|\alpha|^2 + |\beta|^2 = 1$.

(c) (3 marks) Show that this state exhibits sub-Poissonian statistics, i.e. $\mathrm{Var}(n) < \langle n \rangle$.

(d) (2 marks) Briefly describe the significance of sub-Poissonian statistics in light.